Periodic Trends and Properties of Elements

Experimental Overview

The purpose of this experiment is to identify periodic trends in reactivity and solubility of some alkali metals (Group 1), and some alkaline earth metals (Group 2). You will also test for trends in Period 3 (row 3).

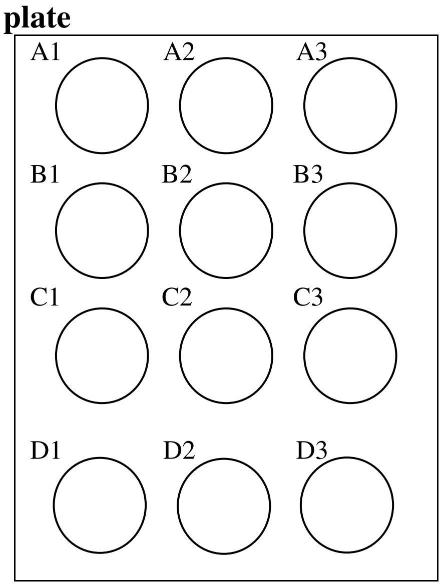
Materials:

- 1 piece each Li, Na, K
- 2 pieces Ca
- 2 pieces Mg ribbon
- 1 piece of Al foil (1cm x 2cm)
- 1 M HCl solution
- forceps
- thermometer
- black & white paper (1 ea)
- red and blue litmus paper
- wooden splint and lighter
- watch glass
- 250 mL beaker

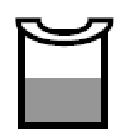
Safety:

Put on goggles. Protective eyewear must be worn at all times due to the dangerous nature of handling alkali metals and their by-products. Never touch an alkali metal. If you get any metal on your skin remove it immediately, flush the area with water, and then wash with soap and water. Since these metals react with water, caution should be used if you are flushing the area. Use copious amounts of water; if any metal remains, a large volume will be needed to neutralize the heat produced when the metal reacts.

Well plate



PART A / Procedure - Reactivity of Metals with Water



1. Half-fill a 250 mL beaker with tap water. Cover the top of the beaker with a watch glass →
2. Get another watch glass, a blade, lit candle, splint, forceps, litmus paper (3 red, 3 blue).
3. Onto a clean, **<u>dry</u>** watch glass place <u>one</u> piece of Li, Na, or K using your forceps (2-3 mm^3 pieces will be cut for you). Observe a small piece of metal and record your observations.
4. Lift the watch glass off of the beaker just high enough to drop the Li into the water. Drop all of the lithium in at the same time. Quickly replace the cover and record your observations. Get ready for step 5.
5. As soon as the metal has stopped reacting, simultaneously lift one end of the watch glass and place a flaming splint in the mouth of the beaker (hold firmly to the watch glass). Record your observations.
6. Touch one piece of red litmus and one piece of blue litmus to the water. Place used litmus in the trash.
7. Dump the water from the beaker into the sink. Carefully rinse the sides of the beaker and the bottom of the watch glass (occasionally metal will splatter onto these surfaces). Repeat for the other 2 alkali metals.
8. Examine pieces of Ca, Mg and Al. Record your observations.
9. Add 20 drops water to wells A1, A2 & A3. Test the water in each well with **<u>red litmus paper</u>** and record initial color. **Litmus is red in neutral or acidic solutions and blue in basic solutions**.
10. Use forceps to add one piece Ca to well A1, one piece Mg ribbon to well A2, and roll half of the 1cm x 2 cm piece of Al into loose ball and add to well A3. Test each well with litmus again and record final color. Change in color means a chemical reaction has occurred.

DISPOSAL OF WASTES: Place all waste in the waste beaker

PART B / Procedure - Reactivity of Metals with Acid (HCl)

1. Add 10 drops of 1M HCl to each of wells C1, C2 & C3.
2. Use your thermometer to measure temperature of acid in one of the wells **before** adding the metals.
   Temperature before adding metals? ______________
3. Put thermometer in each well **before** adding metal. Use forceps to add one piece of calcium to well C1 & record new temp. Repeat for well C2 before adding 1 piece of magnesium ribbon. Roll remaining piece of aluminum foil into a loose ball, and repeat for well C3 before adding aluminum.

DISPOSAL OF WASTES: **Place all waste in the waste beaker**

Data Table C

	Na_2CO_3	Na_2SO_4	KIO_3
$MgCl_2$	A1	A2	A3
$CaCl_2$	B1	B2	B3
$SrCl_2$	C1	C2	C3
$BaCl_2$	D1	D2	D3

Analysis Questions

Questions based on **Parts A and B** -

1. Write the metals tested in Part A from most active to least active.
2. Imagine two electrons in an atom: one is close to the nucleus and one is far away. Which electron should be easier to remove from the atom? Why?
3. Essentially, alkali metals react when an electron in the outer shell is removed. Based on this information, and your answers to 2 and 3, explain why _____ was most reactive and _____ was least reactive.
4. All alkali metals in this lab, reacted according to the equation: metal + water → metal hydroxide + hydrogen. What did we use to test for the presence of metal hydroxides? What did we use to test for hydrogen?
5. Why did the K beaker not react (pop) when the flaming splint was used?
6. Based on your observations, what is the periodic trend in metal activity going down a group of the periodic table?
7. Based on your observations, what is the periodic trend in metal activity across a period of the periodic table going left to right?
8. Look at strontium (Sr) on the periodic table. Would you expect it to be more or less reactive than calcium? Explain.
9. Aluminum is a metal commonly used in making electrical wiring, and structural materials. Why is aluminum used and not magnesium or calcium?